

The **Bode diagram** is a useful tool to represent a frequency response in a graph using a pwl (piecewise linear) model that asymptotically approximates the real plot. There are two types of plots, the **amplitude** and the **phase** plot, both in logarithmic scale.

Why log scale?

We know that $W(jw)$ can be represented in trigonometric form as $W(jw) = |W(jw)|e^{j\arg(W(jw))}$. The, using the neperian complex logarithmic we end up with an expression of form $a + jb$:

$$\ln(W(jw)) = \ln(|W(jw)|) + \ln(e^{j\arg(W(jw))}) = \ln(|W(jw)|) + j \cdot \arg(W(jw))$$

Moreover we use the **bilogarithmic scale**, that is: $20 \log_{10}()$

Recall how a transfer function is usually expressed:

$$W(s) = \frac{n(s)}{d(s)} = \frac{\sum_{m=0}^{\deg(n(s))} b_m s^m}{\sum_{n=0}^{\deg(d(s))} a_n s^n}$$

which can be rewritten as

$$W(s) = \frac{b_m}{a_n} \frac{(s - z_1) \dots (s - z_m)}{(s - p_1) \dots (s - p_n)}$$

With some more algebraic manipulation, and by considering the *Fourier transform* (i.e. the phase representation) it is possible to show that **constant terms, zeroes, poles (real and complex)** give different types of contribution to the bode plot, but all these **contributions are additive**

Remark (Electronics).

In electronics we will have that $\frac{b_m}{a_n} = (\frac{\omega_0}{s})^r$ which includes the r poles in zero and has $(\omega_0)^r = (\frac{b_m}{a_n})^r$

From now on $\log_{10}$ will be simply referred as $\log$

Remark (Semilog scale vs log scale).

Wolfram alpha, and many other representations are in **semilogarithmic scale**. The x axis is in w , not $\log w$.

Notice that in log scale the slope becomes $[dB/\text{decade}^2] \xrightarrow{\log \text{ scale}} [dB/\text{decade}]$.

I will write all results in semilog scale. However the plots are in semilogarithmic scale in order to show more of the graph

Remark.

One final remark regards the bode form. Usually every pole is expressed in the form $1 + s/w$ if instead it is written as $1 - s/w$ then the phase will have the opposite contribution.

In the case of a second degree expression check if it is a pair of real poles and rewrite them. If it is the case of complex poles then the phase will be different only if the first degree term has a negative factor.

0.2) Constant Term

The first term (K_B) is called **Evans gain** and it is important since it is constant, and it also **determines the gain/attenuation**. A simple way to find it is by finding the **DC/Static gain**:

Definition 1 (DC/Static Gain).

Given a transfer function $W(s)$ with no pole in the origin, we define its gain as the value in $W(0) = K_B$. Then we can group this term in the expression.

This can also be extended to the case with some poles in the origin, by just ignoring the term $\frac{w_0}{s}$ in the calculation

Suppose we have

$$W(s) = K_B$$

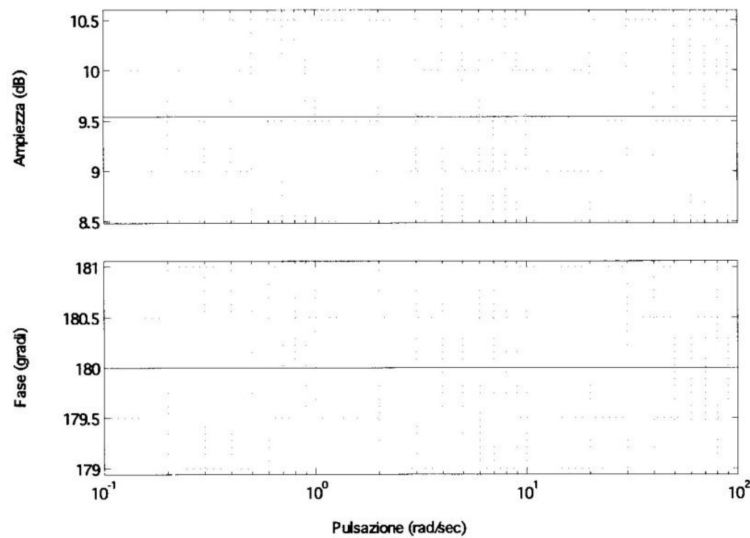
AMPLITUDE PLOT

It is just a straight line of form $y = 20 \log |K_B|$

If $|K_B| > 1$ we have a gain in the output, otherwise we have an attenuation

PHASE PLOT

Since $K_B \in \mathbb{R}$ we have that if $K_B > 0$ the phase is null, otherwise it is $\pm 180^\circ$ it doesn't matter which one



Bode diagram of $W(s) = -3$

0.3) Pole in Origin

Suppose we have

$$W(s) = \frac{1}{s^v}$$

where v is the degree which coincides with the amount of poles in the origin.

AMPLITUDE PLOT

The transfer function will be $|W(jw)|_{dB} = -20v \log w$

This means that the resulting plot will be a line passing trough the origin $[0, 0]$ and it will:

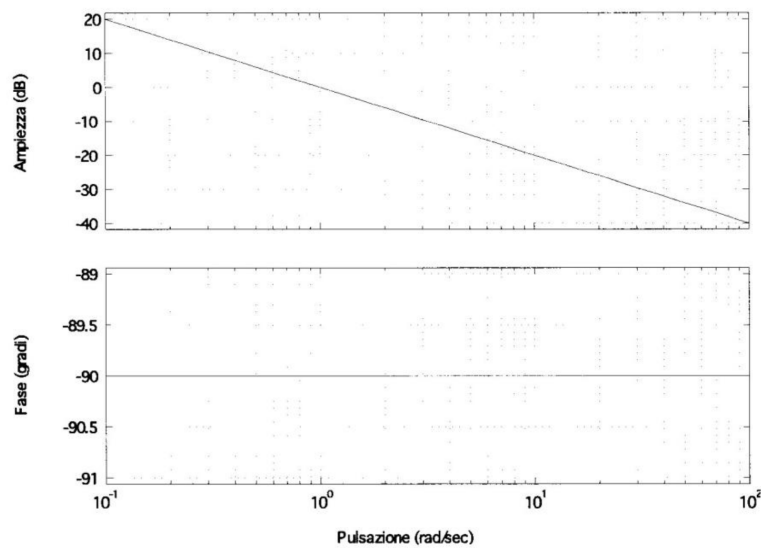
- $v > 0$ go downwards with slope $-20|v|[dB/decade]$
- $v < 0$ go upwards with slope $20|v|[dB/decade]$

PHASE PLOT

The transfer function will be $\arg(W(jw)) = -v \cdot 90^\circ$.

This means that the resulting line will be of form $y = \mp 90^\circ$

- $v > 0 \implies -|v|90^\circ$
- $v < 0 \implies |v|90^\circ$



Bode plot of $W(s)=1/s$

Remark.

A zero in the origin is the equivalent of a pole in the origin, just with different exponential term

Corollary 2 (Pole in Zero with Gain).

As explained before in electronics we will have $(\frac{\omega_0}{s})^r$ as the first term which includes the r poles in zero and has $(\omega_0)^r = (K_B)^r$.

*This is actually useful since you can draw just **one line with the contribution of the zero and constant term included**.*

If we have poles in the origin ($r > 0$) this line has negative slope ($-20|r|[dB/decade]$) and will pass trough $[|r| \log \omega_0, 0]$.

Otherwise, if there are zeros in the origin ($r < 0$) the line has positive slope ($20|r|[dB/decade]$) and will still pass trough $[|r| \log \omega_0, 0]$.

Notice that the lines change sign in the slope and it would suggest that the intersection points would also be switched, but ω_0 becomes it's reciprocal with $r < 0$ and thus the effects cancel out.

The phase plot depends on the sign of $(w_0)^r$.

If it is positive it will give zero contribution to the phase and thus the phase plot will be identical to the case of a pole in the origin.

Otherwise, the phase will be shifted by a factor of 180° .

0.4) Real Poles and Zeroes

Suppose we have

$$W(s) = (1 + sT)^\mu$$

This is valid for zeroes if $\mu > 0$, otherwise it is valid for poles, and $|\mu|$ is the number of poles/zeroes. Both cases are centered in $-1/T$, then we end up with:

$$\begin{aligned}\log |W(jw)| &= \mu \log \sqrt{1 + (wT)^2} \\ \arg(W(jw)) &= \mu \arctan(wT)\end{aligned}$$

Definition 3 (Cutoff frequency).

We define the **cutoff frequency** as

$$w_c = \frac{1}{|T|}$$

This frequency is the minimum required frequency to not have a 0 in the Bode diagram approximation

We can rewrite all the poles and zeroes as $s - p \rightarrow 1 + \frac{s}{p} \rightarrow 1 + Ts = 1 + \frac{s}{w_c}$

I will write the results according to w_c

Here we see the first implementation of the pwl model, since we will consider the following 2 approximations:

$|wT| \ll 1, |wT| \gg 1 \implies w \ll 1/|T|, w \gg 1/|T|$.

Then we have the two cases:

- $w \ll w_c$

$$\log |W(jw)| \approx \mu \log 1 = 0$$

$$\arg(W(jw)) \approx \mu \arg 1 = 0^\circ$$

- $w \gg w_c$

$$\log |W(jw)| \approx \mu \log wT = \mu(\log w + \log T)$$

$$\arg(W(jw)) \approx \mu \arg 1 = \mu \cdot \text{sign}(T) \cdot 90^\circ$$

From here it is clear that **real zeroes and poles behave in the same way, but with inverted sign.**

AMPLITUDE PLOT

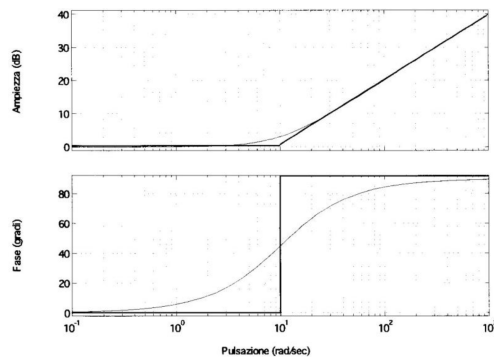
It will be $y = 0$ until point $[\log w_c, 0]$ (cutoff frequency) on the x-axis, then it will be a line going upwards (zero) or downwards (pole) with slope $\pm 20|\mu|[dB/\text{decade}]$.

Remark (Zero in origin).

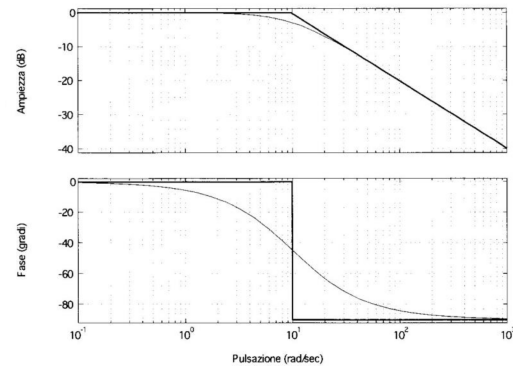
From here we can see that usually the slope goes upwards for zeroes, and downwards for poles. You can also apply this reasoning for Zeroes and Poles in the origin.

PHASE PLOT

The phase will be 0 until point $[\log(w_c) - 1, 0]$, then it will be a straight line connecting to the point $[\log(w_c) + 1, \pm|\mu| \cdot 90^\circ]$ if it is zero or pole.



Bode diagram of $W(s)=1+0,1s$



Bode diagram of $W(s)=(1+0,1s)^{-1}$

The phase plot has a wrong approximation, but from the real curve it is evident how the slope should be

0.5) Complex Conjugate Poles and Zeroes

Suppose we have

$$W(s) = \left(1 + 2\epsilon \frac{s}{w_n} + \left(\frac{s}{w_n} \right)^2 \right)^\mu$$

Now we have:

$$|W(jw)|_{dB} = 10\mu \log \left[\left(1 - \left(\frac{w}{w_n} \right)^2 \right)^2 + \left(2\epsilon \frac{w}{w_n} \right)^2 \right]$$

$$\arg(W(jw)) = \begin{cases} \arctan \left(\frac{2\epsilon \frac{w}{w_n}}{1 - \frac{w^2}{w_n^2}} \right), & w < w_n \\ \text{sign}(\epsilon) \cdot \frac{\pi}{2}, & w = w_n \\ \arctan \left(\frac{2\epsilon \frac{w}{w_n}}{1 - \frac{w^2}{w_n^2}} \right) + \text{sign}(\epsilon) \cdot \pi, & w > w_n \end{cases}$$

Now we can find the asymptotic approximations for $w \ll w_n$, $w \gg w_n$:

- $w \ll w_n$

$$\log |W(jw)| \approx \mu \log 1 = 0$$

$$\arg(W(jw)) = \mu \arg 1 = 0^\circ$$

- $w \gg w_n$

$$\log |W(jw)| \approx 2\mu \log\left(\frac{w}{w_n}\right) = 2\mu(\log w - \log w_n)$$

$$\arg(W(jw)) = \mu \cdot \text{sign}(\epsilon) \cdot \pi = \mu \cdot \text{sign}(\epsilon) \cdot 180^\circ$$

So we have find another case of **cutoff**.

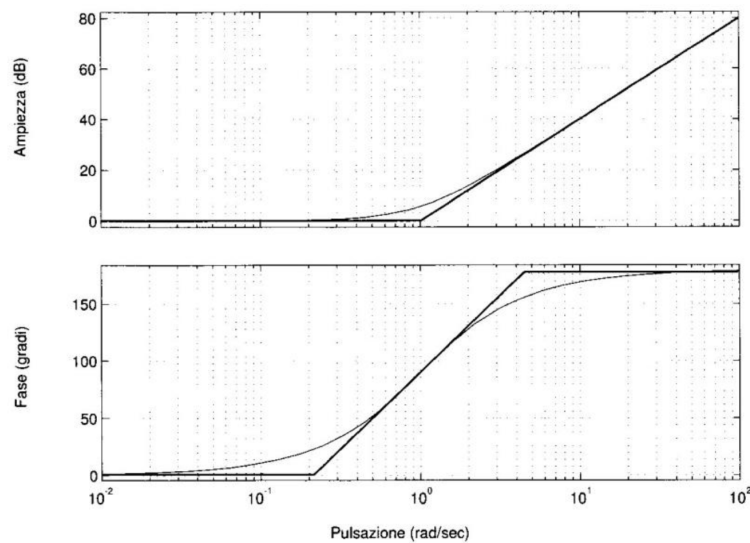
Again we have that μ shows the number of poles or zeroes

AMPLITUDE PLOT

It will be zero until $[\log w_n, 0]$, then it will be a line going upwards (zero) or downwards (pole) with slope $\pm 40|\mu|[\text{dB/decade}]$.

PHASE PLOT

The phase will be 0 until point $[\log(w_c) - 1, 0]$, then it will be a straight line connecting to the point $[\log(w_c) + 1, \pm |\mu| \cdot 180^\circ]$ if it is zero or pole.



Bode plot of $W(s)=s^2+1.9s+1$

Here the approximation of the phase is correct. Moreover in the example we have $w_n = 1$, $\epsilon = 0.95$, $\mu = 1$

0.6) How to Draw a Bode Diagram With Multiples Zeroes and Poles

It's actually pretty easy by recalling the **additive** property that was given at the start

AMPLITUDE PLOT

1. Draw every single line corresponding to every zero/pole/constant term
2. In the resulting plot, every point will have a slope corresponding to the sum of the slopes of every line in that point
3. add the constant value gain/attenuation

Example.

This is a very dumb example, but consider

$$W(s) = \frac{1-s}{1-s}$$

This is clearly 1. And the bode plot will be a straight line of form $y = 0$. But suppose we are a dumb student who doesn't notice this and applies every rule.

1. Zero in 1, that means it gives 0 contribution until $[1, 0]$ and then rises with slope $+20[\text{dB/decade}^2]$

2. Pole in 1, that is 0 contribution until $[1, 0]$ and then goes downwards with slope $-20[dB/decade^2]$
3. Until $[1, 0]$ there is no contribution, then after that point the 2 contributions are $\pm 20[dB/decade^2]$ and thus $20 - 20 = 0[dB/decade^2]$

Applying the rules gives the same result of the "direct" computation $\square$

PHASE PLOT

1. Draw every single line corresponding to every zero/pole/constant term
2. In the resulting plot, every point will have a slope corresponding to the sum of the slopes of every line in that point
3. When the phase has a constant attribution add\subtract it